Stress the budget.
Explain the impact.
Explore the fiscal outlook.

An interactive application that turns a structured budget workbook into financial scenarios and annual fiscal paths. Change property-tax, payroll and enrollment assumptions; inspect the deficit, coverage and capacity to absorb it.

One workbook. Two core analytical tools.

Scenario Analysis shows the immediate financial effect of a selected combination of assumptions. Multi-Year Outlook compounds annual changes and evaluates the timing of balance and reserve sufficiency.

Workflow	What happens
PREPARE	Extract public PDF figures, clean and reconcile them, then populate the Excel template. Python can support this upstream work.
LOAD	Keep the expected seven sheets, metric labels and ratio structure. The dashboard validates mapped inputs and reconciles totals.
EXPLORE	Test tax growth, workforce / compensation and enrollment; explain the operating-result bridge and enter future-year paths.



Scenarios → Sensitivity → Multi-Year Outlook

Explicit assumptions translated into financial consequences

The same preparation workflow can support another district.

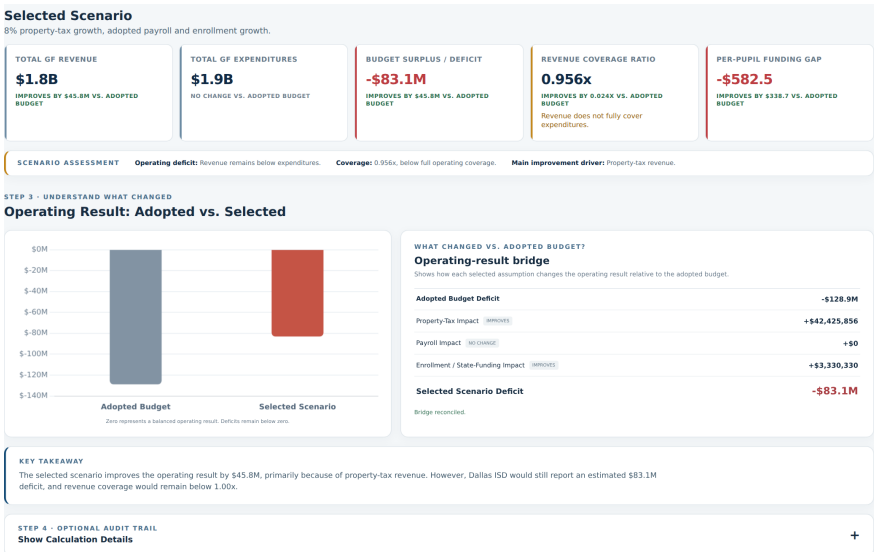
PDF → structured Excel → dashboard. A compatible workbook can supply another district's inputs, but the sheet mapping, tax calibration and state-funding proxy must be checked before interpreting its results. Keeping the same sheet names alone does not establish model suitability.

Data and formulas stay visible.

Excel preserves source data and its ratios. The application recalculates scenario outcomes from mapped financial inputs; it does not execute Excel formulas. All scenarios use the same calculation engine.

Delivery: local standalone HTML, interactive charts and documented methodology. No backend or statistical forecasting model.

Which assumptions change the financial gap?

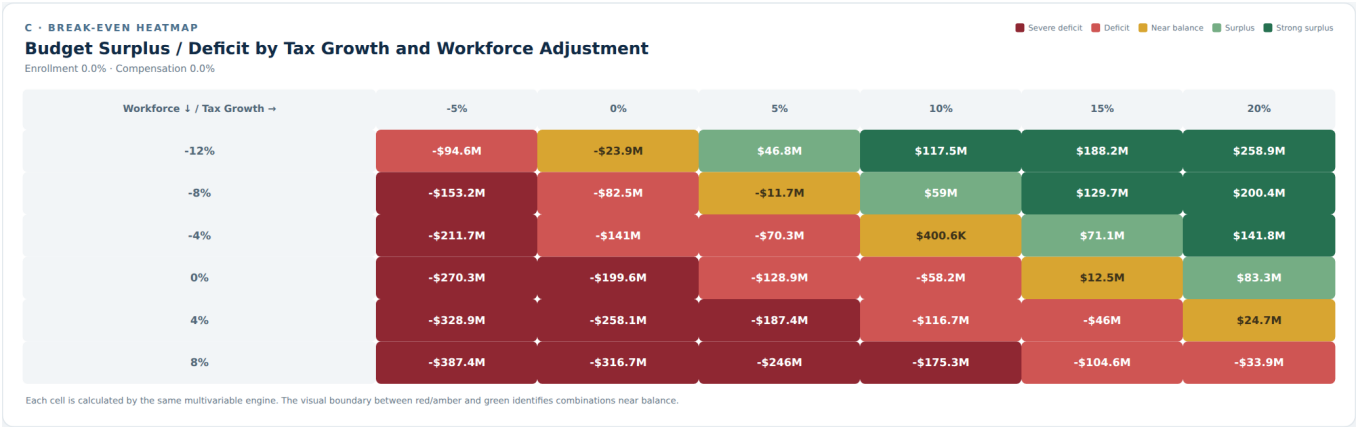


Actual dashboard: selected scenario and operating-result attribution.

Upside is meaningful. The deficit still remains.

8% tax growth and 2% enrollment growth, with payroll unchanged, improve the operating result by $45.8M. The FY2026 adopted deficit of $128.9M becomes $83.1M: $42.4M of tax impact plus $3.3M of modeled state revenue.

Map combinations, rather than one isolated shard.



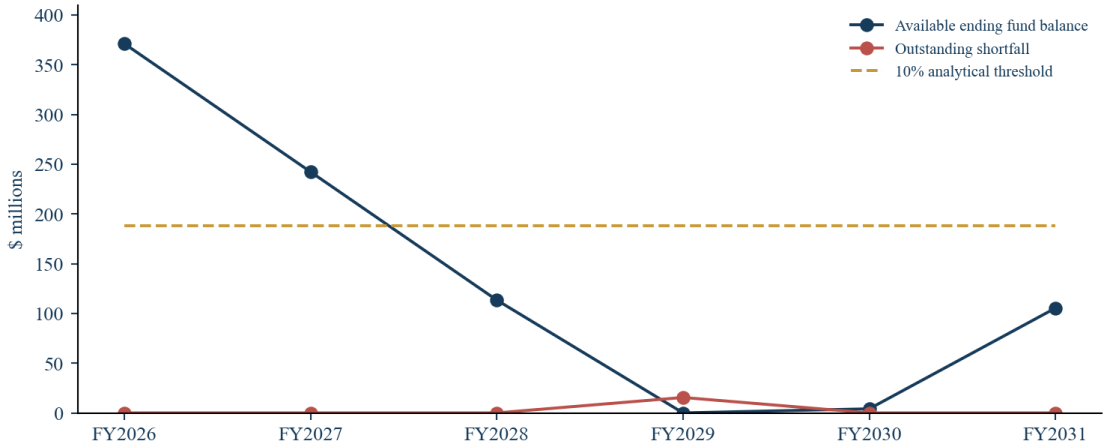
Actual break-even sensitivity matrix: each cell is the budget surplus or deficit for a tax-growth / workforce combination, with enrollment and compensation fixed at 0%. Colors classify the operating result; they are not credit ratings or probabilities.

How the engine works: tax revenue uses a base inferred from adopted tax revenue and the 5% calibration. Payroll = adopted payroll × (1 + workforce change) × (1 + compensation growth). State revenue follows enrollment at a fixed starting revenue per pupil.

Operating result = revenue − expenditures. Revenue coverage = revenue ÷ expenditures. Other-local revenue, federal revenue and non-payroll expenditures remain fixed.

When does balance return? Do reserves last until then?

Start from the adopted budget or selected scenario. Enter 3, 5 or 7 future years of tax growth, workforce, compensation and enrollment changes. Future assumptions compound from the preceding year, rather than repeating a constant deficit.



Illustrative reserve path: $500M at the start of FY2026; tax growth 0% in FY2027–FY2029, 10% in FY2030 and 5% in FY2031; payroll and enrollment changes 0%.

FY2030 balance does not mean reserves survived the path.

Reserves are depleted before recovery. The FY2030 surplus of $19.6M first repays $15.5M of outstanding shortfall; $4.1M remains to rebuild available fund balance. The $500M starting reserve is illustrative, not an official district balance.

Outlook measure	Meaning
First balanced year	First annual operating result ≥ 0.
Sustained recovery	First of two consecutive balanced years with coverage ≥ 1.00x.
Reserve reconciliation	Net position = beginning available balance − beginning shortfall + operating result. Available balance and outstanding shortfall cannot both be positive.

Inspect the assumptions. Use the outputs for discussion.

User-entered paths are not independently validated. State revenue uses a simplified per-pupil proxy; fixed categories do not grow. Fund balance is not cash liquidity, and modeled shortfall is not automatically debt. Results are scenarios, not forecasts or rating opinions.

Project access & implementation

Excel + JavaScript + SheetJS + Chart.js; Vite standalone build. Delivery record: 30 passing unit tests and browser / standalone checks. Full financial methodology report supplied separately. GitHub and hosted demo links pending approved publication.

Source: Dallas ISD FY2025–2026 Adopted Budget Book, pp. 31, 35 and 83. Scenario outputs calculated by the project engine.